Dr. Dipankar Bhattacharya, Ph.D.

Marie Skłodowska-Curie Fellow, Imperial College London

Learning-based Manipulation and Control

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Summary

Robotics researcher with a Ph.D. in Mechatronics Engineering and 5+ years of postdoctoral and industry-facing research at Imperial College London, TransGP (Hong Kong), HKU and CUHK. I build learning-based and model-based systems for robotic manipulation, with focus on deformable objects (imitation learning, mesh-based perception, dual-arm control), and I take them from simulation (Isaac Sim, PyDrake) to real hardware. Previously Senior Research Engineer on an industry-partnered garment-automation project, where I led a multidisciplinary team and translated industrial requirements into deployed manipulation pipelines. Currently an EU-funded Marie Skłodowska-Curie Fellow (EUR 276k) leading my own research programme. **Seeking:** Senior Research Scientist (full-time) or internship; available from June 2027 (internship earlier).

Experience

Marie Skłodowska-Curie Postdoctoral Fellow, Imperial College London, UK **2025 – Present**

CASREx: cable-driven soft exosuit for upper-arm rehabilitation (Horizon Europe grant 101205678, EUR 276,187.92).

- Lead an independent project from grant proposal to hardware: control, learning, simulation and human trials.
- Developing an Isaac Sim digital twin for reinforcement-learning policy training and sim-to-real transfer.
- Imitation learning with a safety filter (SFIL) for series-elastic cable arms under co-contraction; EMG-in-the-loop adaptive control.
- Supervise two MEng placement students; co-supervise PhD candidate Chuanyun Ouyang (University of Auckland); mentor pre-college robotics student Bobomurod Fazliddinov for college applications.
- GTA for Maths 2 (Dyson School); guest lecturer on robotics simulation and control (PyDrake).

Senior Research Engineer – Control Systems, Center for Transformative Garment Production (TransGP), Hong Kong **2024 – 2025**

Mentor: Prof. Kazuhiro Kosuge. Industry-partnered dual-arm fabric manipulation for garment manufacturing.

- Led a multidisciplinary team on dual-arm fabric flattening, destacking and pre-sewing handling (mesh perception + imitation learning); mentored PhD students.
- Built and validated robotic testbeds for industrial sewing and destacking workflows; translated industrial requirements into research deliverables.
- Results: IEEE IROS 2026 (RTFF flattening policy), IEEE RA-L 2025 (automated straight-line sewing), IEEE T-ASE 2026 (fabric alignment via global-local weighted ICP), IEEE ICMA 2025 (fabric destacking).
- Concurrently Visiting Research Associate, Dept. of EEE, Hong Kong University.

Postdoctoral Fellow, The Chinese University of Hong Kong **2021 – 2024**

Supervisor: Prof. Darwin Lau. Modeling and control of cable-driven robots.

- Kinematic/dynamic modeling of cable–object interference and wrapping; validated in simulation and hardware (Mech. Mach. Theory 2025).
- Iterative-learning-based reactive control for redundant multilink/hybrid cable robots (IEEE T-CST 2023).
- Designed a cable-driven growing vine robot for ceiling inspection; co-authored two ITF grants; mentored students.
- Visiting Postdoc, LS2N, École Centrale de Nantes (Oct 2021 – Jan 2022): iterative-learning control experiments.

Research Assistant / Graduate Teaching Assistant, The University of Auckland, New Zealand **2017 – 2020**

- Industry-funded stent-testing research (BvW Holding, NZD 39,000) on a robotic soft esophagus; taught four undergraduate courses.

Lecturer (Assistant Professor Grade-1), Galgotias University, India **2013 – 2016**

Education

Ph.D., Mechatronics Engineering, The University of Auckland (2016–2021). *Robotic Soft Esophagus: Design, Modelling, and Control.* Cited in *Nature Communications* (2026).

M.Tech, Systems and Control, IIT Roorkee (2011–2013), GPA 8.4/10. **B.Tech, Electronics and Communication Eng.**, NERIST (2004–2010), GPA 9.1/10.

Selected Publications

- **D. Bhattacharya***, K. Tang*, et al., "RTFF: Random-to-Target Fabric Flattening Policy Using Dual-Arm Manipulator," *IEEE/RSJ IROS*, 2026. (*equal)
- W. Wang, **D. Bhattacharya**, et al., "Robotic Fabric Alignment System for Sewing Using Global-Local Weighted ICP," *IEEE T-ASE*, 2026 (accepted).
- B. Jin*, A. Kobayashi*, **D. Bhattacharya**, et al., "Automated Straight-line Sewing of Stretchable Fabrics with Different Lengths," *IEEE RA-L*, 2025.
- **D. Bhattacharya**, T. K. Cheung, Y. Wang, D. Lau, "Kinematic and Dynamic Modeling of Cable-Object Interference and Wrapping in Complex Geometrical-Shaped CDPRs," *Mech. Mach. Theory*, 214, 2025.
- **D. Bhattacharya**, et al., "Tri-Space Operational Control of Redundant Multilink and Hybrid CDPRs Using an Iterative-Learning-based Reactive Approach," *IEEE T-CST*, 2023.
- **D. Bhattacharya**, L. K. Cheng, W. Xu, "Nonlinear Model Predictive Control of a Robotic Soft Esophagus," *IEEE T-IE*, 2021.
- **D. Bhattacharya**, et al., "RoSE: A Robotic Soft Esophagus for Endoprosthetic Stent Testing," *Soft Robotics*, 2021.
- **D. Bhattacharya**, L. K. Cheng, W. Xu, "Sparse Machine Learning Discovery of Dynamic Differential Equation of an Esophageal Swallowing Robot," *IEEE T-IE*, 67(6), 2020.

Full list (15+ journal and conference papers): Google Scholar; code and project websites: bhattner143.github.io/dips

Skills

Learning and control: imitation learning, reinforcement learning, sparse system identification, nonlinear MPC, iterative-learning control, mesh-based deep perception.
Software: Python, PyTorch, C/C++, MATLAB/Simulink, ROS2, Isaac Sim, PyDrake, LeRobot, OpenSim, Git, Linux.
Hardware: dual-arm manipulators, cable-driven actuation, NVIDIA Jetson, Raspberry Pi, myRIO, Arduino; CAD in SOLIDWORKS/Onshape.

Grants, Awards and Service

- Marie Skłodowska-Curie Postdoctoral Fellowship, Horizon Europe (2025, EUR 276k); co-author on two Hong Kong ITF grants (2022, 2023); BvW Holding industry grant (2018, NZD 39k).
- Selected, European Talent Academy 2026; nominee, ECRI Creating a Positive Research Environment Award (2026); Best Conference Paper, RiTA 2017.
- Invited talks: Harvard University (2026), Northeastern University (2026), World Robotics Conference Beijing (2025). Presenter at IEEE IROS 2026, ICRA 2023/2024.
- Reviewer: IEEE T-ASE, RA-L, ICRA; ASME J. Mechanisms and Robotics. IEEE member.